Sandesh Joshi

AI Software Engineer | Backend Developer | IoT & Embedded Systems

Ajman, UAE |

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PROFESSIONAL SUMMARY

AI engineer with backend and IoT experience. Develops AI algorithms and software that use large amounts of data: RAG document retrieval (PostgreSQL, pgvector), OpenAI API integrations and CNN computer vision, served through Python, FastAPI and Django APIs. Evaluates model effectiveness, collects and analyzes sensor and application data, and works with teams to set project goals. Delivered measurable results: 90% less manual news writing, 81% aircraft classification accuracy and a reported 50% cut in emergency response and preparation time. Completed an eight-month AI/ML microdegree and presented research at ICET 2025 to stay current with AI developments.

TECHNICAL SKILLS

AI & Machine Learning: Machine Learning, AI Algorithms, Model Evaluation (accuracy), Retrieval-Augmented Generation (RAG), OpenAI API, LLM Integration, Computer Vision, Convolutional Neural Networks (CNN), AI Document Retrieval, Vector Search (pgvector), Speech-to-Text, Text-to-Speech

Backend & APIs: Python, FastAPI, Django, Node.js, REST API Development, API Integration, Backend Development, Full-Stack Development

Data & Infrastructure: Data Collection, Data Analysis, PostgreSQL, pgvector, SQL, Query Optimization, Cloud Dashboards, Team Collaboration

IoT & Embedded Systems: ESP32, ESP8266, Arduino, Sensors and Actuators, IR/RF Modules, GSM, Wi-Fi Connectivity, Cloud Dashboards, UAV/Drone Systems, Flight Controllers, Embedded C/C++ Firmware

Languages & Frontend: Python, C/C++ (embedded), JavaScript, TypeScript, PHP, SQL, React.js, Next.js, HTML5, CSS3, SCSS

PROFESSIONAL EXPERIENCE

Full-Stack & IoT Developer | RedMug IT Solution L.L.C | Dubai, UAE (On-site) May 2026 – Present

- Engineered client web applications and backend services in Python, Django and JavaScript, integrating REST APIs across internal and third-party systems
- Optimized database queries and backend logic to improve application performance and reliability
- Built automation tools and IoT integrations that stream device data to live web dashboards

Hardware & IoT Project Engineer | Nepal Innovation Center (NIC) | Nepal 2022 – 2024

- Designed and tested ESP32, ESP8266 and Arduino sensor prototypes that sent live data over Wi-Fi to cloud dashboards for remote monitoring and control
- Integrated UAV sensors and payloads, configured flight controllers and supported drone field testing
- Developed IR/RF modules and wrote embedded C/C++ firmware for microcontroller-based devices

PROJECTS

EvidenceAI – RAG Document Research Platform | Python, FastAPI, Next.js, TypeScript, PostgreSQL, pgvector | GitHub

- Developing a full-stack platform that answers questions over uploaded documents with source-linked citations
- Implementing pgvector-based retrieval, private workspaces and saved research conversations

Poker News Generator & Unified AI Services | Python, FastAPI, OpenAI API

- Contributed to an OpenAI-powered news generator that cut manual news writing by 90%
- Contributed to a FastAPI service for speech-to-text, text-to-speech and text/image generation, with cost-aware model selection using live pricing and performance data

Aviation Safety Classifier & NIC RAG Chatbot | Computer Vision, CNN, RAG

- Contributed to a CNN aircraft fit-to-fly classifier that achieved 81% accuracy
- Contributed to a RAG chatbot that automates responses for NIC Nepal

LOC-8 – Accident Detection & Vehicle Tracking | Next.js, Python, ESP32, GSM, Machine Learning

- Contributed ML and software for vehicle tracking, accident detection and geofenced emergency alerts; reported a 50% reduction in response and preparation time

Smart-Farmer – IoT Farmer Assistance System | Python, FastAPI, Next.js, IoT sensors

- Contributed ML and software that link pH, NPK and humidity sensor readings to crop recommendations for farmers

Additional Projects

- **Accident Detection System** (final-year project, GitHub): IoT and embedded system combining sensors, microcontrollers and embedded C/C++ for accident detection and monitoring
- **Drone Development:** integrated flight controllers, sensors and payloads; participated in UAV field testing
- **Pokhara Ride, Pokhara Sarathi, corporate websites** (RedMug IT Solution, Grace International): contributed ride-sharing and community platform features and built responsive sites with React.js, Next.js, TypeScript, Node.js and PHP

EDUCATION

Bachelor of Computer Engineering | Pokhara University | Pokhara, Nepal Jan 2021 – Jun 2026

- CGPA 3.00/4.00; coursework completed, final result awaited. Final-year project: Accident Detection System

Higher Secondary (+2 Science, PCM + CS) | National Academy of Science and Technology (NAST) | NEB, Nepal 2019 – 2021

CERTIFICATIONS & RESEARCH

- **AI/ML Microdegree**, AI Community in Nepal (AICN), Oct 2025 – Jun 2026, Certificate No. AIML-MD-2026-021: eight-month program in ML theory, implementation and a capstone project; credentials: Machine Learning Scientist, Data Scientist Associate
- **Research paper presenter**, ICET 2025, Pokhara Engineering College: quantum causal inference and two-bit quantum image generation

ACHIEVEMENTS & LEADERSHIP

- Second Runner-Up, Hackathon Nova (72-hour), Pokhara Engineering College, 2025; participant, Code For Change Ideathon (2025) and IDEA TANK, NAST (2023)
- Admin Lead, Code for Change Pokhara (2023–2024): organized workshops and mentorship reaching 1,000+ students

LANGUAGES

Nepali (Native) | Hindi (Fluent) | English (Advanced)